

R&D DOCUMENT ON THE CREATION OF AN INTERNAL AND EXTERNAL LOAD BALANCER

PREPARED BY :

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PURPOSE :

To explore an Azure load balancer and create an internal and external load balancer, and verify if it's working

AZURE LOAD BALANCER OVERVIEW

- Azure Load Balancer is a cloud service that distributes incoming network traffic across backend virtual machines (VMs) or virtual machine scale sets (VMSS).
- Load balancing refers to the efficient distribution of incoming network traffic across a group of backend virtual machines.
- It operates at layer 4 of the OSI model.
- It serves as the single point of contact for clients.
- Azure Load Balancer has three stock-keeping units (SKUs) - Basic, Standard, and Gateway. Each SKU is catered towards a specific scenario and has differences in scale, features, and pricing.

CORE CAPABILITIES

- High availability for the distribution of resources
- Scalability
- Low Latency by using pass-through load balancing
- Flexibility
- Health Monitoring

DEPLOYMENT OF LOAD BALANCER - EXTERNAL

1. Create a NAT Gateway

Microsoft Azure

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Create network address translation (NAT) gateway

...

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Basics

Outbound IP

Subnet

Tags

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Azure NAT gateway can be used to translate outbound flows from a virtual network to the public internet.
[Learn more about NAT gateways.](#)

Project details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

azure subscription

Resource group *

(New) load-balancer-rg

[Create new](#)

Instance details

NAT gateway name *

lb-nat-gateway

✓

Region *

East US

▼

Availability zone ⓘ

No Zone

▼

TCP idle timeout (minutes) * ⓘ

15

✓

4-120

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2. Create a Virtual Network

>>

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Create virtual network ...

**Basics**[Security](#)[IP addresses](#)[Tags](#)[Review + create](#)

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.

[Learn more.](#)**Project details**

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Subscription *

azure subscription



Resource group *

(New) load-balancer-rg

[Create new](#)**Instance details**

Virtual network name *

lb-vnet

Region ⓘ *

(US) East US

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Edit subnet



Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

IP address space ⓘ

10.0.0.0/16



10.0.0.0 - 10.0.255.255 (65536 addresses)

Subnet details

Subnet template ⓘ

Default



Name * ⓘ

backend-subnet

Starting address * ⓘ

10.0.0.0

Subnet size ⓘ

/24 (256 addresses)



IP address space ⓘ

10.0.0.0 - 10.0.0.255 (256 addresses)

Security

Simplify internet access for virtual machines by using a network address translation gateway. Filter subnet traffic using a network security group. [Learn more](#)

NAT gateway ⓘ

lb-nat-gateway



[Create new](#)

Network security group ⓘ

None



[Create new](#)

Route table

None



Save

Cancel

3. Create a Load Balancer

Create load balancer ...



Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers uses a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. [Learn more.](#)

Project details

Subscription *

Resource group * [Create new](#)

Instance details

Name * ✓

Region *

SKU * ⓘ ☒ Standard ☐ Gateway ☐ Basic (Retiring on September 30, 2025)

Type * ⓘ ☒ Public ☐ Internal

Tier * ☒ Regional ☐ Global

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4. Testing the Load Balancer



DEPLOYMENT OF LOAD BALANCER - INTERNAL

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Create network address translation (NAT) gateway ...

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Resource group *
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[Create new](#)

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Availability zone ⓘ
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2. Create a Virtual Network



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Create virtual network ...



Basics

Security

IP addresses

Tags

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[Learn more.](#)

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[Create new](#)

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[Create new](#)

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4. Test the Load Balancer



Hello World from myVM1

▲ CONCLUSION

The deployment of internal and external Azure Load Balancers was successful. External Load Balancer ensured internet-facing traffic reached Web Tier VMs, while Internal Load Balancer securely managed traffic between App and DB tiers. Verification confirmed proper traffic distribution and tier isolation, achieving high availability and secure architecture.
